is called voltage. It is represented by the letter 'V' and the unit is volt. It is measured by volt-meter. The other parameters termed as volts are EMF, Potential and Potential Difference.

i) Electro Motive Force (EMF)

It is the force which causes the flow of electrons in any closed circuit. It is represented by volt.

ii) Potential and Potential Difference

The work done in bringing unit positive charge from infinity to that point against the application of electric field is called potential. It is also represented as volt.

The difference of potential between any two points in a electrical circuit is called potential difference and is expressed as volts only.

The following table represents the difference between electromotive force (EMF) and Potential Difference (PD).

Electro Motive Force (EMF)	Potential Difference (PD)
i) EMF refers to source of electricity only.	PD exists between any two points in a circuit.
ii) It is measured when the circuit is open.	It is measured on a closed circuit.
iii) It does not depend upon the resistance of the circuit.	It depends upon the resistance of the circuit and is directly proportional to it.
iv) It is greater than the potential difference in the same circuit.	It is less than the electromotive force.

2.4.3 Electric Power

Power is rate of doing work. The power is obtained by the following expressions.

$P = V \times I$. The unit is watt

P = Power

V = Applied Voltage

I = Current

2.4.4 Resistance

Resistance may be defined as the property of a substance to oppose the flow of current flowing through it. It is represented by the letter R and the unit is ohm(Ω). It is measured by ohm meter. Mega ohms value is measured by using megger

2.4.5 Laws of Resistance

The resistance of a conductor in a circuit depends upon the following:

- It depends upon the material used and its properties
- Resistance value of a conductor is directly proportional to the length of the conductor
- It is inversely proportional to the area of cross section of a conductor
- It also depends upon the temperature of the conductor when the current is flowing through it

2.5 OHM'S LAW

In an electrical circuit the current, voltage, and resistance are related to one another. The relationship was derived by the scientist ***Georg Simon Ohm***. So it is called Ohm's Law.

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